

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY****Third Semester of B. Tech. Examination (CL/ME)****March- April 2018****CL 243 Mechanics of Solids****Date: 28/03/2018, Wednesday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

**SECTION - I****Q - 1 (a) Do as Directed.****[05]**

1) As per Hooke's law, Within elastic limit, Stress is \_\_\_\_\_ proportional to Strain.

a) Directly b) Inversely c) Not

2) Poisson's ratio is ratio of \_\_\_\_\_ (longitudinal to lateral strain, lateral to longitudinal strain, shear stress to shear strain)

3) The \_\_\_\_\_ is found from the stress vs strain relation of a material. (Modulus of Elasticity, Modulus of Rigidity, Bulk Modulus)

4) Define Principle stresses with the help of sketch.

5) On a principle plane the magnitude of shear stress will be equal to \_\_\_\_\_.

[maximum, minimum, zero]

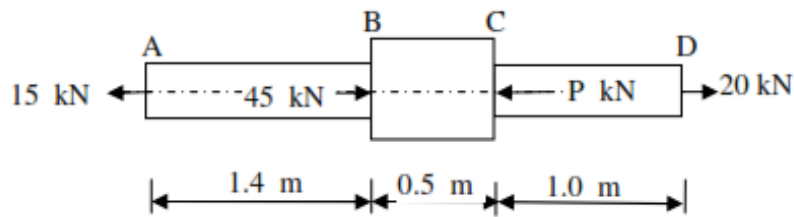
**Q - 1 (b) Define With Sketch (Any Two)****[02]**

1-Shear Stress, 2-Tensile stress, 3-Volumetric strain, 4- Thermal stress

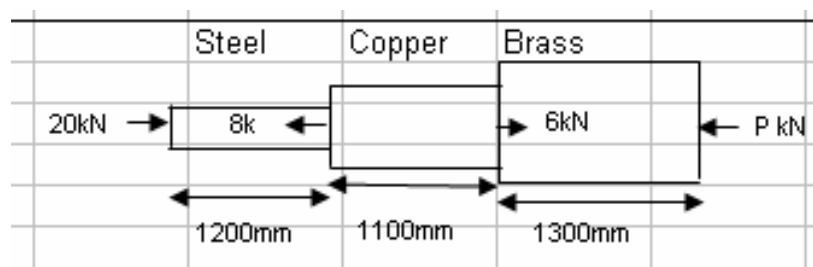
**Q - 2 (a)** A stepped bar ABCD is axially loaded as shown in **fig.** is in equilibrium. Determine (i) Magnitude of unknown force 'P' (ii) stresses in each part and (iii) total change in length of the bar. Details of each part of the bar is as follow:

**[07]**

| Sectional/Material properties | Part AB | Part BC | Part CD |
|-------------------------------|---------|---------|---------|
| C/S Area in mm <sup>2</sup>   | 500     | 600     | 500     |
| Modulus of Elasticity in Gpa  | 210     | 100     | 80      |

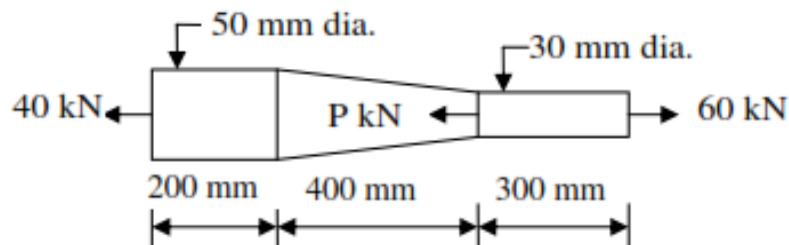


- (b) A stepped bar made of steel, copper and brass is under axial force as shown in **figure** [07] and is in equilibrium. The diameter of steel is 12mm, diameter of copper is 16mm and the diameter of brass is 20mm. Determine (i) Magnitude of unknown force P (ii) stresses in each material and (iii) Total change in length of the bar. Take  $E_{\text{steel}} = 200\text{GPa}$ ,  $E_{\text{copper}} = 100\text{GPa}$  and  $E_{\text{brass}} = 80\text{GPa}$



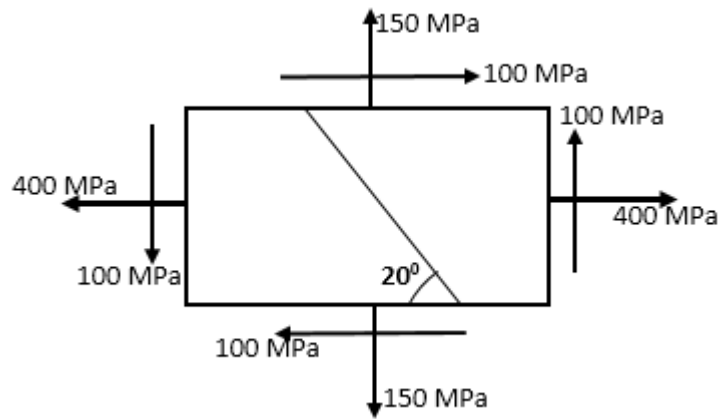
OR

- (b) An assembly of steel bars as shown in the fig. is in equilibrium. Find force P and the net elongation of the assembly. Take  $E = 2 \times 10^5 \text{ N/mm}^2$ . [07]



- Q - 3 (a) Explain Strain Energy due to Gradual loading. [04]  
 (b) Write Any two. [10]

- 1 A plane element is subjected to the stresses as shown in figure below. Find out the (i) Principal stresses and their directions. (ii) Maximum shear stress and their direction. Use **Analytical Method**.



- 2 Solve Q-1(above) using **Graphical Method** and find out (i) Principal stresses and their directions. (ii) Maximum shear stress and their direction.
- 3 A solid rod is of length 2 m and cross-section 100 mm x 10 mm. If the applied axial force in the rod is 50 kN suddenly, Find the strain energy stored and the modulus of resilience. Take  $E = 2 \times 10^5 \text{ N/mm}^2$ .

### SECTION - II

**Q - 4 (a) Do as Directed.**

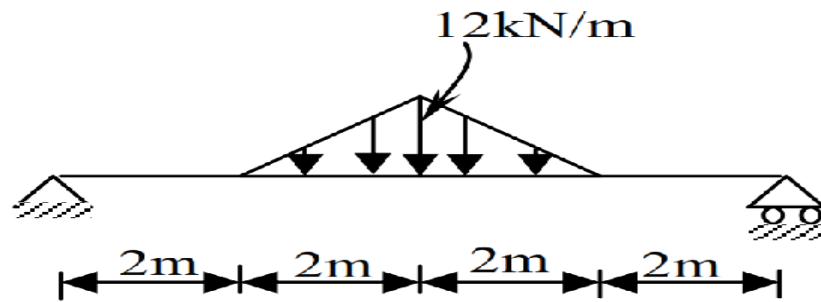
**[07]**

- 1) Beam extends beyond the support then that beam is known As\_\_\_\_\_ beam.
- 2) At free end of a cantilever bending moment is always\_\_\_\_\_ unless a concentrated Moment is applied at the free end.
- 3) A M.S. bar under tension test shows the property of\_\_\_\_\_.  
[malleability, ductility, tension ability]
- 4) Sketch qualitative shear stress distribution diagrams of 'I' section of the beams.
- 5)When shear force at a point is Zero, then bending moment at that point will be\_\_\_\_\_.  
[ Zero, Maximum, Minimum, infinity]
- 6)At neutral axis bending stresses is \_\_\_\_\_.  
[Minimum,Maximun, Zero]
- 7) Moment of inertia of an area is minimum about its \_\_\_\_\_axis.

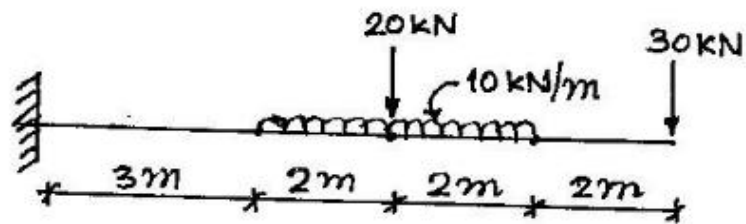
**Q - 5 (a) Write Any two.**

**[14]**

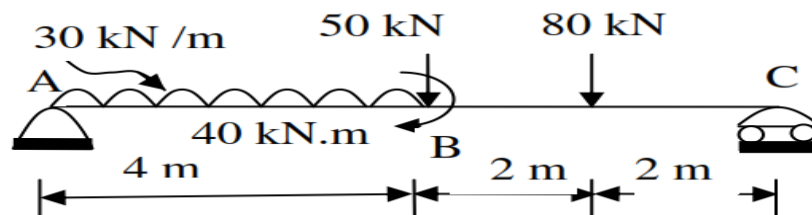
- 1 ) Draw shear force and bending moment diagrams of the beams loaded as Shown in Fig.



2) Draw shear force and bending moment diagrams of the beams loaded as shown in Fig.



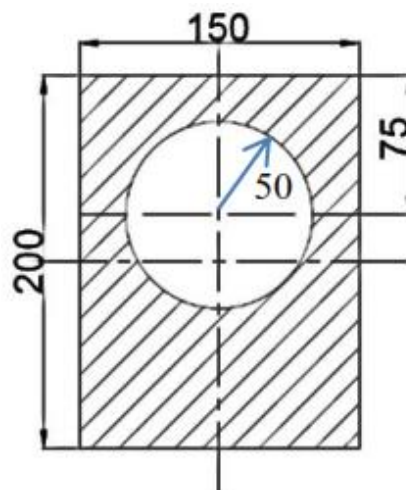
3) Draw shear force and bending moment diagrams of the beams loaded as shown in Fig.



Q - 6 (a) Write Any Two.

[14]

- 1 Determine MI of a hollow section shown in **Figure**. about an axis passing through its Center of gravity. All dimension are in mm,



- 2 Determine the MI of the 'I' section about its central X and Y axis. Size of 'I' section is:  
Top flange:  $100\text{mm} \times 20\text{mm}$ , Bottom flange:  $150\text{mm} \times 20\text{mm}$ , Web:  $20\text{mm} \times 150\text{mm}$ .
- 3 A beam is having cross section of "T" is subjected to vertical shear force of 100kN. Calculate shear stresses at neutral axis and at the junction of flange and web, Draw shear stress distribution diagram. Dimension of flange =  $160\text{mm} \times 20\text{mm}$  , Web=  $140\text{mm} \times 20\text{mm}$ .
- 4 Derive formula for maximum shear stress in a Triangular section. Also sketch shear stress distribution across the section.

\*\*\*\*\*